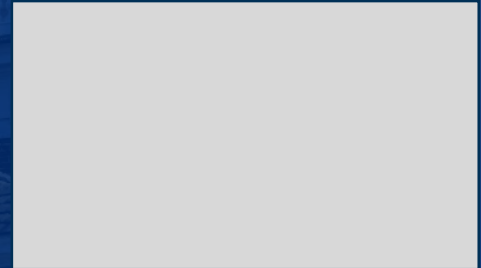


Applied Generative AI

Welcome to the Multimodal RAG



What is Multimodal RAG?

Definition: "Multimodal RAG systems can access and reason over textual and visual data."

Key Features:

- Combines text and images in a unified retrieval process.
- Generates grounded, context-rich answers.

Use Case Example: Diagram-based context retrieval alongside textual explanations.

Traditional RAG System: Recap

Process Overview:

- Text-based query → Encoding → Retrieve text chunks → Large Language Model generates an answer.

Use Case: "Document Q&A, code-based search, knowledge assistant."

Limitations of Text-Only RAG:

"What if the relevant information is in a diagram, image, or schematic?"

Use Case: Complex instructions in visuals.

Fields impacted: Manufacturing, medicine, aerospace, architecture.

Multimodal RAG Pipeline

Stages:

1. Ingestion (Extract text and images from documents)
2. Embedding (Convert text/images to vectors)
3. Indexing (Store in a multimodal database)
4. Retrieval (Unified retrieval of text and images)

Simple pipeline flow:

(Ingestion → Embedding → Indexing → Retrieval).

Example: Robotics Industry

Scenario: Technician asks, "How do I reset the safety sensor?"

Traditional RAG: Retrieves text but misses the critical image.

Multimodal RAG: Retrieves both text and diagram.

Visual of the text and diagram retrieval side by side.

- **LLM Options:** GPT-4V (handles both text and image input).
- Example: Text and image passed to GPT-4V for integrated answer generation.
- Image showing a query and response in a multimodal LLM (text and visual output).

Challenges in Multimodal RAG

- Embedding alignment issues (CLIP vectors not always semantically consistent).
- Latency concerns (slower processing for images).
- Prompting complexity (formatting and tuning needed).
- Evaluation difficulties (human-in-the-loop needed).

Optional Homework Assignment

- Extract text and diagrams from a PDF (technical manual).
- Embed text with sentence transformers and diagrams with CLIP.
- Store and index in Weaviate or QDrant.
- Build retrieval using Langgraph and feed into GPT-4.

Goal: Create a multimodal system that generates coherent responses based on visual and textual input.

Key Takeaways

- Multimodal RAG expands traditional systems by integrating visual, textual, and contextual information.
- This system is powerful for real-world, high-stakes applications like healthcare and manufacturing.
- Understanding both the opportunities and challenges is essential for building multimodal systems.